

14 THOUSAND

THE NUMBER OF SPECTATORS THAT CAN BE SEATED AT THE THEATER IN EPIDAUROS

The acoustic properties of the theater at Epidaurus in Greece, built by Polycleitus the Younger in the 4th century BCE, allowed the actors to be heard perfectly up to 197 ft (60 m) from the stage.

GREEK MEDICINE MADE SIGNIFICANT ADVANCES

in the 4th century BCE after the dissection of human bodies was pioneered by Diocles of Carystus, who wrote the first book devoted to anatomy. The **foundation of the Museum**, a scientific academy set up by Ptolemy I of Egypt (367–283 BCE), helped give rise to an Alexandrian school of medicine. One member, Herophilus of Chalcedon (335–280 BCE), identified the brain as the **seat of the nervous system** and made a distinction between arteries and veins.

Greek understanding of physics also progressed under Strato of Lampsacus (c.335–269 BCE). He rejected the idea of a force pushing light objects, such as air, upward to counter



Via Appia

The first major Roman road, the Via Appia, originally ran from Rome to Capua. It was gravelled; paving stones were added in 295 BCE.

the force that pulls heavy objects down. He argued for the **existence of a vacuum** and showed that, because air can be compressed, voids must exist between the particles of which it is made up.

In Europe, wooden trackways had been used to traverse wet and marshy ground since Neolithic times, but proper roads needed a strong, centralized political authority to build and maintain them. In 312 BCE, the Romans began to construct a **vast network of roads** that bound their empire together. The first road they built, which ran from Rome to Capua, was called Via Appia. Roman roads were 10–26 ft (3–8 m) wide and were laid out on solid clay beds or timber frameworks, filled with loose flint or gravel. Sometimes they were **bound together with lime mortar** and topped with paving stones, or cobblestones in cities.

The Pharos of Alexandria was commissioned c.300 BCE by the ruler of Egypt, Ptolemy I. It was the tallest lighthouse in the ancient world at 410–492 ft (125–150 m) high. **Innovative**

engineering was used in the hydraulic machinery needed to raise fuel to the fire that burned on top by night. During the day, a mirror of polished metal or glass reflected the Sun to create a warning beacon for ships.

Pythagoras had experimented with acoustics in the 6th century BCE. Aristotle advanced his work further in the 4th century BCE by theorizing that **sound consisted of contractions and expansions in the air**. The Greek theater at Epidaurus used stepped rows of seats to filter out low-frequency background noise, which allowed actors to be heard perfectly in the back row.

Compiled before 300 BCE, the Chinese text *Huang Di Nei Jing* explains human physiology and pathology in terms of the balancing forces of the universe: the opposing, but mutually dependent, principles of yin and yang; the five elements (earth, fire, wood, water, and metal); and qi, the essence of which everything is composed. **Ill health was**



Pharos of Alexandria

The Pharos of Alexandria was one of the Seven Wonders of the World. It was destroyed by an earthquake in the 14th century.

thought to be caused by an imbalance of yin and yang, in the patient's qi, and in the five elements that had their counterparts in the organs of the body and the environment.

ARISTOTLE (384–322 BCE)

A founding figure in Western philosophy, Aristotle was a pupil at Plato's Academy in Athens. During his career he wrote more than 150 treatises on almost every aspect of Greek philosophy and science. He taught an empirical approach, that knowledge is gained from experience, and that all matter consists of a changeable form and an unchangeable substance.



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THE NUMBER OF PLATONIC SOLIDS (REGULAR POLYHEDRALS) IN EUCLIDEAN GEOMETRY

